Task6

本任务重新构建了网络，并在原有的基础上增加了新的终端设备，配置了ip地址和网关，使用OSPF协议实现了全网互通。

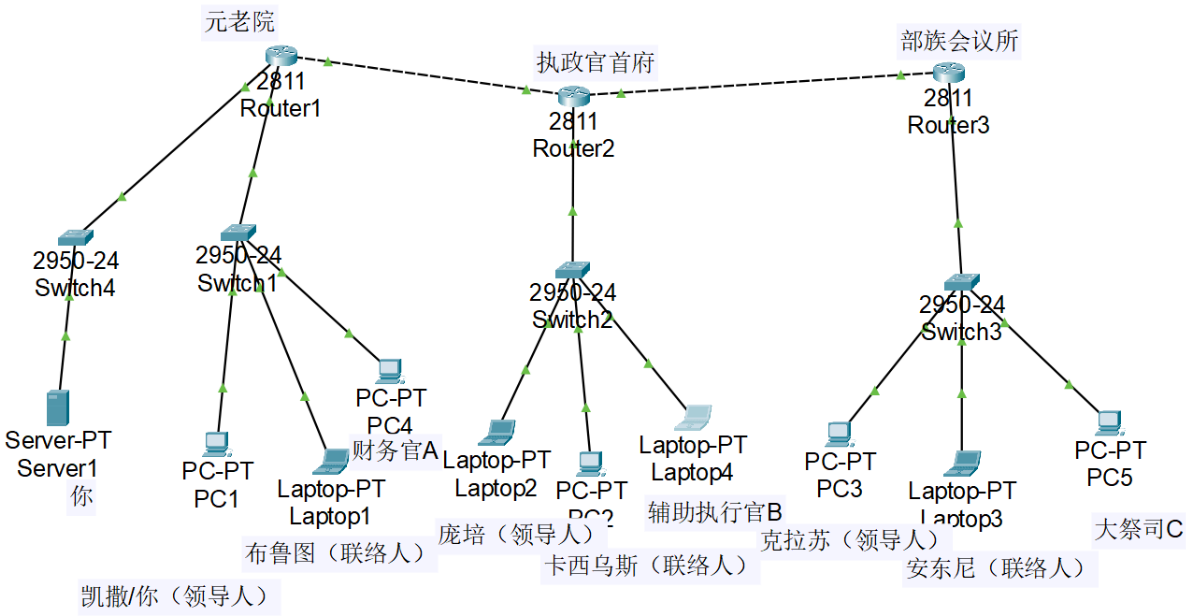
物理隔离 Server1： 由于原拓扑中 Server1 与 PC1 位于同一广播域（同一交换机下），路由器 ACL 无法拦截其二层通信。为了实现对 Server1 的严格访问控制，我在 Router1 旁新增了一台交换机（Switch4），并将 Server1 移至一个新的独立子网（192.168.4.0/24），连接至 Router1 的 `FastEthernet1/0` 接口 。

联络人机制： 只要通信双方有一个联络员就能通信

ACL 策略方向： 配置来自谁的信息可以进入我的子网(in是子网内部想要访问外部，out是子网外部想要访问内部)

为了实现task6，在router1,router2,router3上配置扩展ACL。

Device	Port	IP Address	Gateway	User/Role
Router1	Fa0/0	192.168.1.1/24	-	元老院网关
	Fa0/1	10.0.1.1/24	-	R1-R2互联
Router2	Fa1/0 (或对应的LAN口)	192.168.2.1/24	-	执政官首府网关
	Fa0/0	10.0.1.2/24	-	R2-R1互联
	Fa0/1	10.0.2.2/24	-	R2-R3互联
Router3	Fa0/1	192.168.3.1/24	-	部族会议所网关
	Fa0/0	10.0.2.1/24	-	R3-R2互联
PC1	Nic	192.168.1.2/24	192.168.1.1	凯撒 (元老院领导)
Server1	Nic	192.168.4.2/24	192.168.4.1	机密管理
Laptop1	Nic	192.168.1.4/24	192.168.1.1	布鲁图 (元老院联络人)
PC4	Nic	192.168.1.5/24	192.168.1.1	财务官A
PC2	Nic	192.168.2.2/24	192.168.2.1	卡西乌斯 (执政官联络人)
Laptop2	Nic	192.168.2.3/24	192.168.2.1	庞培 (执政官领导)
Laptop4	Nic	192.168.2.4/24	192.168.2.1	辅助执政官B
PC3	Nic	192.168.3.2/24	192.168.3.1	克拉苏 (部族领导)
Laptop3	Nic	192.168.3.3/24	192.168.3.1	安东尼 (部族联络人)
PC5	Nic	192.168.3.4/24	192.168.3.1	大祭司C



OSPF配置：

r1

```
router ospf 1
 network 192.168.1.0 0.0.0.255 area 0
 network 192.168.4.0 0.0.0.255 area 0
 network 10.0.1.0 0.0.0.255 area 0
```

r2

```
router ospf 1
 network 192.168.2.0 0.0.0.255 area 0
 network 10.0.1.0 0.0.0.255 area 0
 network 10.0.2.0 0.0.0.255 area 0
```

r3

```
router ospf 1
 network 192.168.3.0 0.0.0.255 area 0
 network 10.0.2.0 0.0.0.255 area 0
```

ACL配置：

r1

```
conf t
! --- 办公区保护规则 ---
ip access-list extended SENATE_LAN_OUT
 ! 1. 领导人特权：允许庞培(2.3)和克拉苏(3.2)访问凯撒(1.2)
 permit ip host 192.168.2.3 host 192.168.1.2
 permit ip host 192.168.3.2 host 192.168.1.2
```

```
 ! 2. 回包放行：允许 Server1(4.2) 回复凯撒
 permit ip host 192.168.4.2 host 192.168.1.2
 ! 3. 联络人通道：允许任何人发起对布鲁图(1.4)的访问
 permit ip any host 192.168.1.4
 ! 4. 双向通信保障：允许外部联络人(2.2, 3.3)的流量进入(发信/回信)
 permit ip host 192.168.2.2 any
 permit ip host 192.168.3.3 any
exit

! --- 服务器保护规则 ---
ip access-list extended SERVER_GUARD
 ! 仅允许凯撒访问
 permit ip host 192.168.1.2 host 192.168.4.2
exit

! --- 应用规则 ---
interface FastEthernet0/0
 ip access-group SENATE_LAN_OUT out
interface FastEthernet1/0
 ip access-group SERVER_GUARD out
```

r2:

```
ip access-list extended CONSULATE_LAN_OUT
 ! 1. 领导人特权：允许凯撒(1.2)和克拉苏(3.2)访问庞培(2.3)
 permit ip host 192.168.1.2 host 192.168.2.3
 permit ip host 192.168.3.2 host 192.168.2.3
 ! 2. 联络人通道：允许任何人发起对卡西乌斯(2.2)的访问
 permit ip any host 192.168.2.2
 ! 3. 双向通信保障：允许外部联络人(1.4, 3.3)的流量进入(发信/回信)
 permit ip host 192.168.1.4 any
 permit ip host 192.168.3.3 any
exit

interface FastEthernet1/0
 ip access-group CONSULATE_LAN_OUT out
```

r3

```
ip access-list extended TRIBAL_LAN_OUT
 ! 1. 领导人特权：允许凯撒(1.2)和庞培(2.3)访问克拉苏(3.2)
 permit ip host 192.168.1.2 host 192.168.3.2
 permit ip host 192.168.2.3 host 192.168.3.2
 ! 2. 联络人通道：允许任何人发起对安东尼(3.3)的访问
 permit ip any host 192.168.3.3
 ! 3. 双向通信保障：允许外部联络人(1.4, 2.2)的流量进入
 permit ip host 192.168.1.4 any
 permit ip host 192.168.2.2 any
exit

interface FastEthernet0/1
 ip access-group TRIBAL_LAN_OUT out
```

正确性验证：

1.权力机构内部成员可以相互通信

用laptop2 ping PC2，预期结果：通信成功

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time=2ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

2.在权力机构之间的相互通信只能通过几位联络人进行，例如：执政官首府的成员只能与元老院的联络人布鲁图进行通信。

用PC2去ping Laptop1,预期结果可以通信

```
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=126
Reply from 192.168.1.4: bytes=32 time=2ms TTL=126
Reply from 192.168.1.4: bytes=32 time<1ms TTL=126
Reply from 192.168.1.4: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

用PC2去ping PC1，预期结果可以通信

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=3ms TTL=126
Reply from 192.168.1.2: bytes=32 time=5ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 5ms, Average = 2ms
```

3.三个权力机构的领导人（凯撒、庞培和克拉苏）可以互相通信讨论工作，进行领导人会议。

用Laptop2 ping PC3，预期结果可以通信

```
C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time<1ms TTL=126
Reply from 192.168.3.2: bytes=32 time<1ms TTL=126
Reply from 192.168.3.2: bytes=32 time<1ms TTL=126
Reply from 192.168.3.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

4.Server1只能与PC1通信

用Server1 去 ping Laptop1，预期不能通信

```
Pinging 192.168.1.4 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

用laptop1 去ping server1，预期不能通信

```
C:\>ping 192.168.4.2

Pinging 192.168.4.2 with 32 bytes of data:

Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.

Ping statistics for 192.168.4.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

用Server1去ping PC1，预期可以通信

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

注：该任务中，不能通信指既不能收数据也不能发数据，可以通信指既能收数据也能发数据。

task7

由于ACL是配置谁可以进入该子网，如果配置ACL的话，router2 和router3要允许PC1的流量访问内部网络的所有设备，同时router1的网关要允许所有目的ip为PC1的流量通信，确保凯撒可以收到回信，但是，这样会导致任何设备都可以访问凯撒，这是不正确的

所以要用icmp协议

router2

```
enable
conf t
ip access-list extended CONSULATE_LAN_OUT
 permit ip host 192.168.1.2 any
exit
```

router3

```
enable
conf t
ip access-list extended TRIBAL_LAN_OUT
 permit ip host 192.168.1.2 any
exit
```

router1

> 只允许 ICMP 的“回信” (Echo Reply) 通过

```
enable
conf t
ip access-list extended SENATE_LAN_OUT
permit icmp any host 192.168.1.2 echo-reply
exit
```

结果验证

用PC1去ping laptop4,PC5，都可以跑通

```
C:\>ping 192.168.2.4

Pinging 192.168.2.4 with 32 bytes of data:

Reply from 192.168.2.4: bytes=32 time<1ms TTL=126
Reply from 192.168.2.4: bytes=32 time<1ms TTL=126
Reply from 192.168.2.4: bytes=32 time<1ms TTL=126
Reply from 192.168.2.4: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.3.4

Pinging 192.168.3.4 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.4: bytes=32 time<1ms TTL=125
Reply from 192.168.3.4: bytes=32 time<1ms TTL=125
Reply from 192.168.3.4: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.3.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

用laptop4去pingPC1，不能跑通

```
Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

task8

无法使用静态路由的原因：在网络搬迁后，元老院 (Router1) 与执政官首府 (Router2) 之间隔着公网 (Router4)。

私网地址不可路由：内部网络使用的是 RFC1918 私有 IP 地址（192.168.x.x），这些地址在公网路由器（Router4）上没有路由表项，且公网路由器通常会丢弃私网地址的流量。

安全性：直接通过静态路由转发会导致数据在公网上明文传输，容易泄露机密 。

解决方案：必须建立 IPSec VPN 隧道，将私网数据包封装在公网 IP 包中进行加密传输。

先断开 Router1 与 Router2 的直接连接。

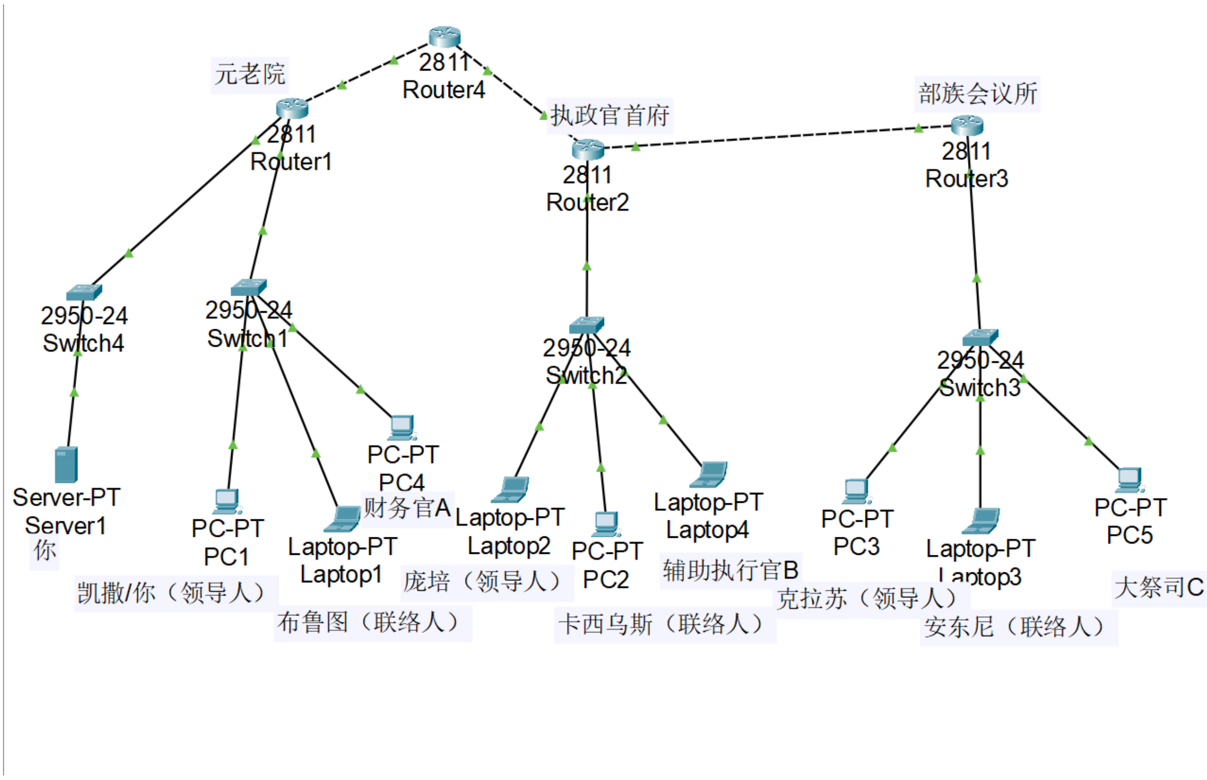
加入 Router4 模拟公网环境。

配置router的新ip：

Router1 (Fa0/1): 10.0.1.1/24

Router4: Fa0/0 (10.0.1.2), Fa0/1 (10.0.3.1)

Router2 (Fa0/0): 10.0.3.2/24



删除原有的OSPF进程

Router1

```
enable
conf t
no router ospf 1
exit
```

Router2

```
enable
conf t
no router ospf 1
exit
```

Router3

```
enable
conf t
no router ospf 1
exit
```

既然删了 OSPF，Router3 现在是个孤岛，要用静态路由把r2和r3连起来。

在 Router2 上 (指向 R3):

```
! 假设 R2-R3 互联 IP 网段是 10.0.2.0/24，R3 是 10.0.2.1
! 告诉 R2：要去 192.168.3.0 (部族)，请找 R3
ip route 192.168.3.0 255.255.255.0 10.0.2.1
```

在 Router3 上 (指向 R2):

```
ip route 0.0.0.0 0.0.0.0 10.0.2.2
```

IPSec VPN 详细配置:

router1:

```
conf t

crypto isakmp policy 10
 encryption 3des
 hash md5
 authentication pre-share
 group 5
exit

! 配置密钥，对端是 R2 的公网 IP (10.0.3.2)
crypto isakmp key cisco123 address 10.0.3.2

! 数据加密
crypto ipsec transform-set VPN_SET esp-3des esp-md5-hmac

! 定义ACL
! 凡是 元老院(1.0, 4.0) 发往 执政官/部族(2.0, 3.0) 的，走 VPN
ip access-list extended VPN_TRAFFIC
 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
 permit ip 192.168.4.0 0.0.0.255 192.168.2.0 0.0.0.255
 permit ip 192.168.4.0 0.0.0.255 192.168.3.0 0.0.0.255
exit

! 创建 Crypto Map
crypto map VPN_MAP 10 ipsec-isakmp
 set peer 10.0.3.2
 set transform-set VPN_SET
 match address VPN_TRAFFIC
exit

! 应用到公网接口
interface FastEthernet0/1
 crypto map VPN_MAP
exit
```

router2

```
conf t

crypto isakmp policy 10
 encryption 3des
 hash md5
 authentication pre-share
 group 5
exit

crypto isakmp key cisco123 address 10.0.1.1

crypto ipsec transform-set VPN_SET esp-3des esp-md5-hmac

ip access-list extended VPN_TRAFFIC
 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
 permit ip 192.168.2.0 0.0.0.255 192.168.4.0 0.0.0.255
 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255
 permit ip 192.168.3.0 0.0.0.255 192.168.4.0 0.0.0.255
exit

crypto map VPN_MAP 10 ipsec-isakmp
 set peer 10.0.1.1
 set transform-set VPN_SET
 match address VPN_TRAFFIC
exit

interface FastEthernet0/0
 crypto map VPN_MAP
exit
```

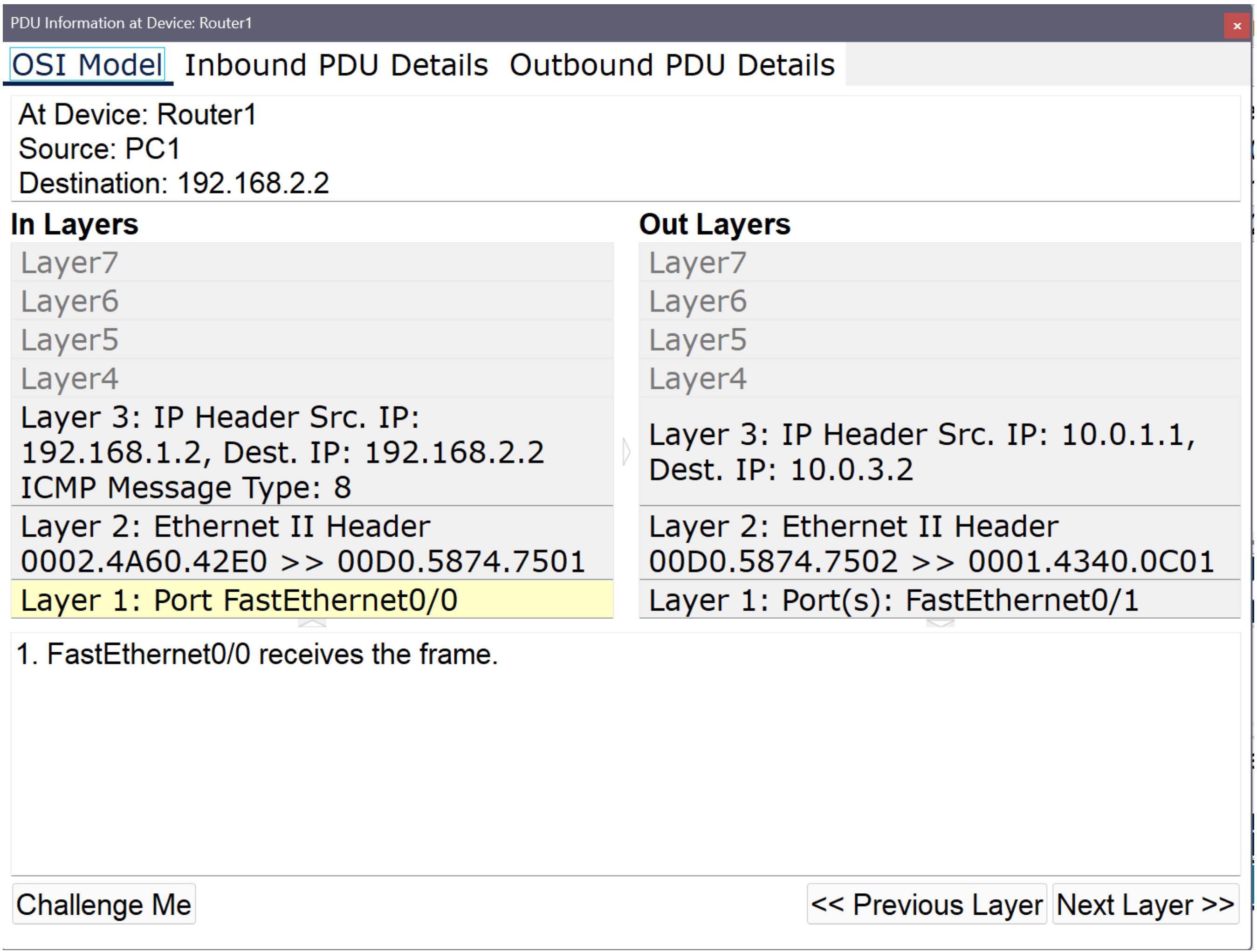
验证:

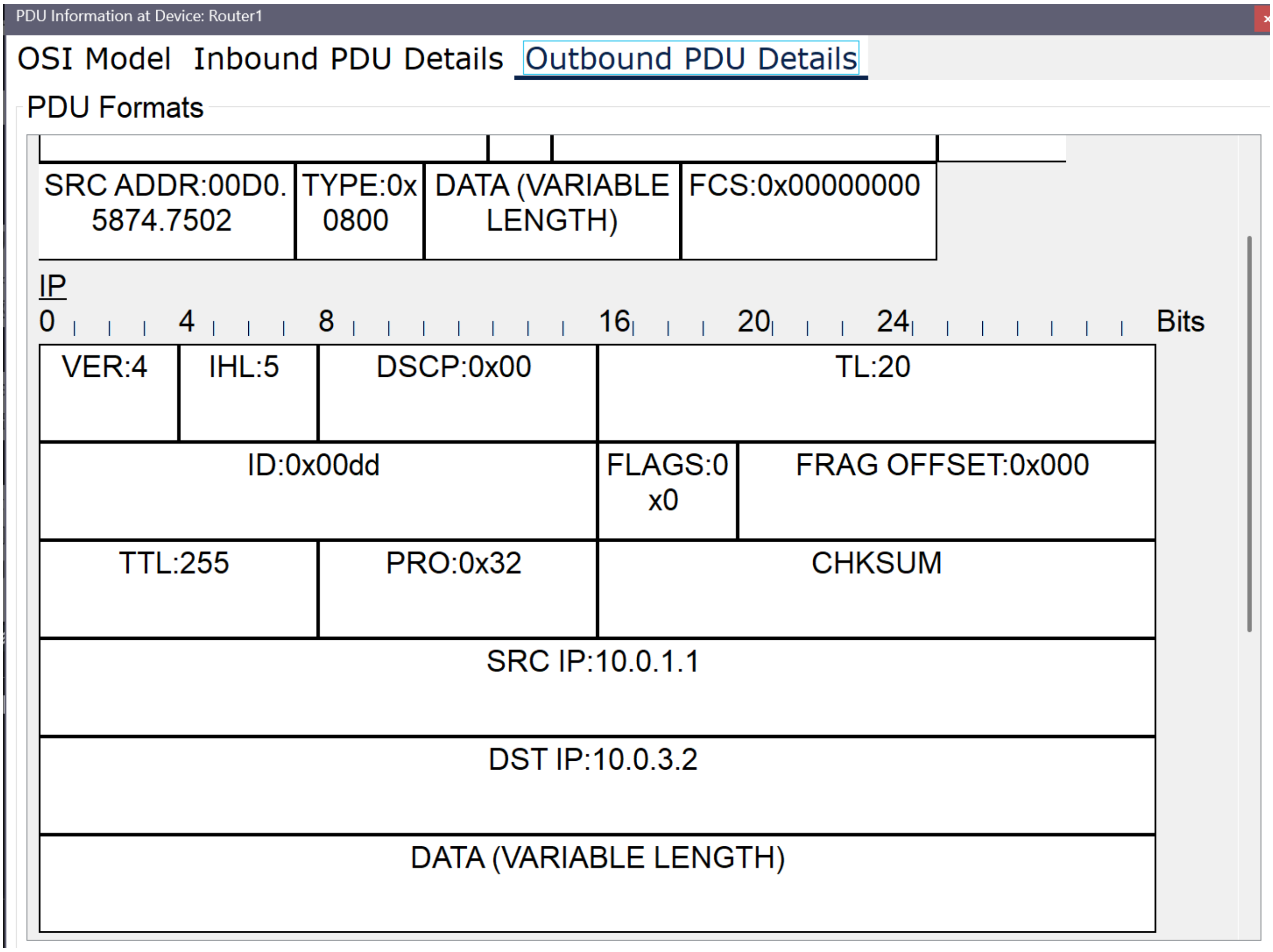
用PC1去ping PC2，预期可以跑通

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

抓包分析

在 Packet Tracer 的 Simulation 模式下，观察 Router1 发往 Router4 的数据包详情（Outbound PDU）。可以看到数据包结构发生了明显变化：





内层 IP 头：源地址为 `192.168.1.2`，目的地址为 `192.168.2.2`。这是原始的私网通信数据。

外层 IP 头：在原始 IP 头之外，新增了一个 IP 头。源地址为 `10.0.1.1`（Router1 公网接口），目的地址为 `10.0.3.2`（Router2 公网接口）。

因为IP Header发生改变，所以本次配置的 IPSec VPN 使用的是隧道模式

原因：

隧道模式会将整个原始 IP 数据包进行加密和认证，并将其封装在一个新的 IP 数据包中。新的 IP 头使用 VPN 网关（即 Router1 和 Router2）的公网地址进行路由。

传输模式只加密数据载荷（Payload），保留原始 IP 头。这要求原始 IP 地址在公网中是可路由的，通常用于端到端通信。

本实验是 Site-to-Site VPN 场景，通信双方（192.168.x.x）使用的是私有 IP 地址，在公网（Router4）中不可路由。因此，必须使用隧道模式将私网数据包封装在公网 IP 包中，才能穿越互联网到达对端网关。